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Xenakis in the modern day

Iannis Xenakis was a Greek-Romanian composer praised for his unique approach to the world of composition and music. Xenakis had adopted a new philosophy to his compositional practice, caring for how he can represent intelligence and complexity through his music.

Xenakis was one of the innovators of what he called “Stochastic” music, utilizing probability distributions and other mathematical methods in his music. This fought against much of the rising popularity of serialism, a compositional technique that had taken over the world of composition at the time. Xenakis, in his article “A Critique of Serialism”, argued that serialism was abstract for the sake of being abstract, falsely portraying a seemingly perceived randomness by the listener, whereas in reality the music was completely deterministic ^[1], giving every choice to the composer. Serialists such as Boulez who composed pieces that applied serialism to every musical parameter available to them, resulting in what was known as total or integral serialism.

Xenakis felt that serialism lacked the mathematical foundation behind its composition, creating an unwanted discrepancy between the complexity of the ear and the music.^[2]

With the aid of computers and various programming languages, this process of mathematical composition has become significantly easier. Through this paper we will be exploring the importance of Xenakis' work, and how it can be used to approach modern composition through software such as Max in various forms and mediums.

Pieces such as *Achorripsis* are able to highlight this musical complexity that Xenakis intended. Xenakis was able to write the full compositions using the poisson distribution, applying it to pre-determined parameters with an arbitrary mean. In the case of *Achorripsis*, these parameters were timbre, duration, frequency and intensity. Through applying these distributions to all of these parameters and through other modulations, full bodies of work are able to be produced.^[3]

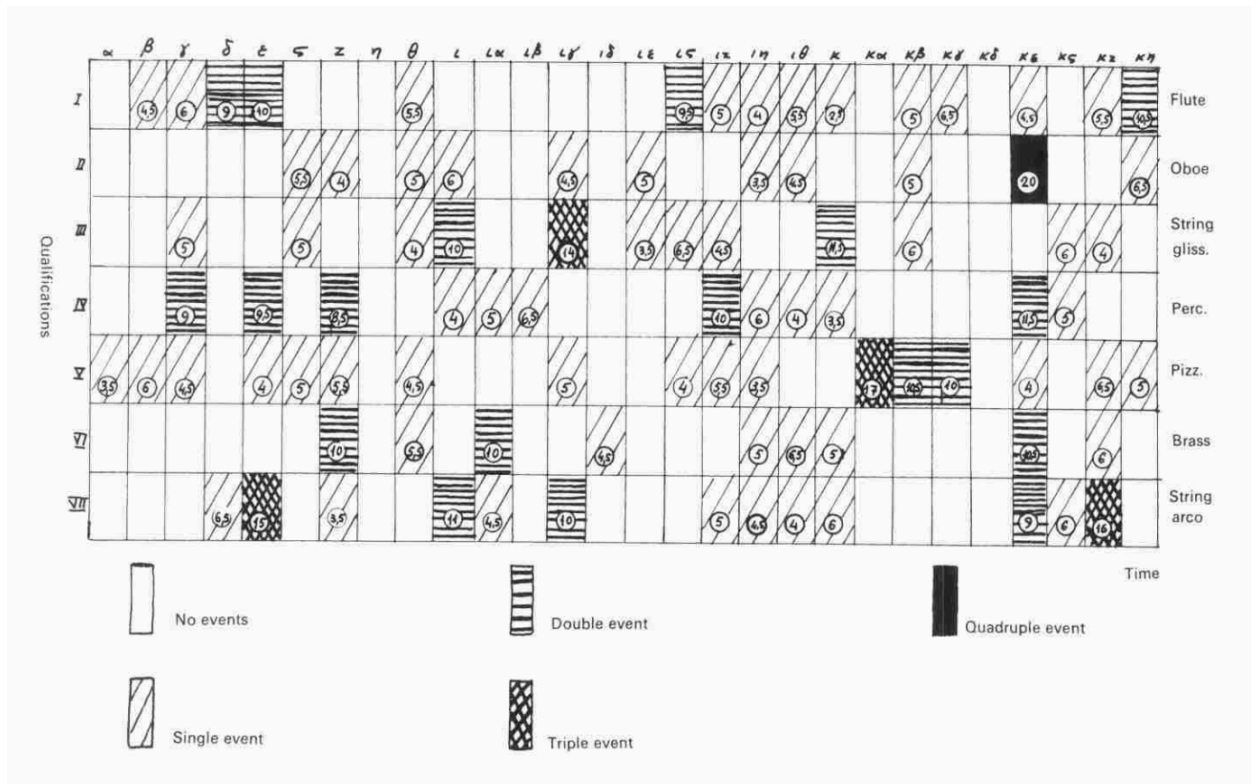
When approaching the topic of Xenakis and his compositions, it is important to understand the foundations of his work. Without knowing how to listen to or analyse the music, there is much left to be desired. What risks being left behind can often provide more musical fulfillment than listening to the piece itself. Stochastic composition often demands the listener to look beyond pitch class and timbre, but rather structure and the various processes through which the music was made. Only then is the listener able to grasp and understand the music, once they have found something to follow.

Achorripsis, as previously mentioned, utilized poisson distributions to fuel the compositional process of the piece. However, the poisson distribution is not the sole function that results in the frequency, timbre and duration of notes, as the distribution is not able to provide this without being applied. These distributions are then used as probability variables to determine these parameters. In the case of pitch class, for example, Xenakis used this distribution to determine the interval between the new note

and the previous note, later on being split off into instruments with the appropriate range in the specific timbre category^[4]. Xenakis was very particular about his instrumentation for each timbre, trying to enable the most range of pitch possible for each section.

The parameter Xenakis labelled as timbre was used to determine the density of notes, as well as what instrument group played them. By using the poisson distribution once again to create a probability distribution of the number of events per cell, it is possible to calculate the number of times each number of events will happen, such as how many cells will one event happen. To then place these on the matrix, another poisson distribution was used. By taking the number of times each event will happen, and dividing that by the number of units of time, you can create an average, which can be used to recalculate the distribution across the columns. This then needs to be redistributed across the rows, following a similar pattern of calculating averages, and creating new probability distributions for each event^{[2][3]}.

The structure of the piece itself can be visualized by a 28x7 matrix, representing 7 timbres and 28 units of time.



[2]

With the aid of this graphical representation, the progression of timbre and density presents itself in a more structured manner. Each column represents one unit of time, 15 seconds. Based on this matrix we can see that the first 15 seconds of the piece feature pizzicato strings, later being joined by what is called the flute section, which contains the piccolo and two clarinets ^[4].

While the piece can be written using pure math, Xenakis often found himself making slight adjustments to the music, either for ease of calculation or because it simply sounded better to him.^[4] Xenakis considers this part of the process incredibly important, as it gives the composer a more direct impact on the piece, letting their ideas reflect in specific patterns the listener might recognise. In Xenakis' own words:

“...during successive rehearsals the relations between the events of the sample ordained by "chance" will form a network, which will take on a definite meaning in the mind of the listener, and will initiate a special" logic," a new cohesion capable of satisfying his intellect as well as his aesthetic sense; that is, if the artist has a certain flair. “

These “networks” are up to the composer to imagine and create, either through visual aid or the music itself, but it is important to be explored and applied^[2]. *Achorripsis* was met with mixed reviews, receiving harsh responses from the French during its performance in 1959 by the Lamoreaux Orchestra. This however did not stop Leonard Bernstein from conducting the piece in 1962 in New York.^[2]

In 1962, Iannis Xenakis created the program “The Free Stochastic Music Program”. This program allowed users to create their own stochastic compositions, by setting parameters for certain conditions, at which point the software generates probability distributions and then sequences that the user can manipulate^[6]. This was used to five pieces: *ST/48, 1-240162, ST/10, 1-080262 for chamber ensemble, ST/4, 1-080262 for string quartet, Morsima-Amorsima (ST/4, 2-030762) for piano, violin, cello, and bass, and Atrées (ST/10, 3-060962) for chamber ensemble*. This program applied the same principles used for *Achorripsis*, using probability distributions, running on FORTRAN II before being upgraded to FORTRAN IV.

The Free Stochastic Music Program had to run on a multi-million dollar computer, at IBM France. Without the ability to perform in real time, we lost the ability to

generatively compose music. With the power of Max and various other music programming languages. It also allows us to parameterize significantly more attributes of not only music and audio, but video as well. The following is an example of this being applied in a video installation I have composed in November 2025.

With the possibilities of control in Max, the first step was to decide what aspects of the composition to be controlled. As this was a multichannel installation, the first parameter chosen was the speaker amplitudes. As the composition didn't involve actual instruments, but rather samples of audio, the other parameter selected was the density of sounds within the unit of time. In order to create distribution tables that can be used to manipulate these parameters, limits need to be set, along with an average value per unit of time, the λ . In Xenakis' words, these numbers are arbitrary, often chosen by him for ease of calculation^[4]. These values can be set to variables making them easy to change when experimenting with what different results can be produced through the manipulation of them.

What came next was finding a way to create probability distributions in Max. In a Poisson Distribution, the probability of an event happening k amount of times in a time period with the average number of times the event happening is λ can be calculated using this formula:

$$\frac{\lambda^k e^{-\lambda}}{k!}.$$

Using a combination of [uzi] and [counter] objects, we are able to create loops in Max, allowing us to easily iterate through values in these calculations. The first important part to implement is the factorial, which is done using a decreasing for loop, multiplying the value of a counter each time it gets triggered by the [uzi] object. After

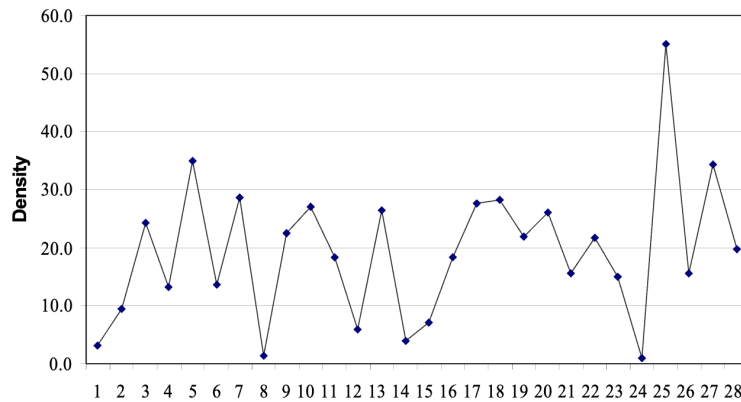
that, creating a function for generating the probability distributions required two parameters, a maximum number of occurrences to calculate up to, and a λ value. With these two inputs a probability distribution is generated and iterated through using [iter] to be used in other processes.

For the triggering of samples, the distribution table generated is used to generate a binary sequence every time a new unit of time begins (this is done by outputting triggers using a metronome). The iterator receives each probability for 0 - k events to occur, generates a random number and compares it to the probability and then sends a 0 or 1 to an accumulator that outputs the result of the last k inputs. For each success (1) in the sequence, the number of samples being triggered is spread evenly across the beat. For example, if a metronome at 60bpm trigger resulted in the output of 001001, the resulting output would be two samples played at a 500ms interval, and 5 samples played at an interval of 200ms.

At the same trigger interval, a similar sequence is generated for four speakers. To allow for two different levels of volume, two poisson distributions are simultaneously created with λ s of 1 and 3. A success in each speaker results in half its maximum gain in increase, only allowing for maximum gain if both speakers succeed.

With these two samples, we gain immense control over a real time stochastic program. With every parameter being free to be changed in real time, create a huge variety of textures and sounds from the same system by just changing the λ of one distribution. With that much control. With a way of generating curves and lines, with an object such as [function] we can create curves that control these parameters as they

progress through time, similarly to how Xenakis diagrammed his changes of density for *Achorripsis*.



Note that in our implementation, we are increasing the probabilities of events happening, resulting in a larger density. These changes could then again be parameterized and randomized by another use of the poisson distribution, resulting in a generative piece of dynamically changing densities. However it is important to consider at which point is the composer's influence recognizable or necessary beyond the initial creation of the piece. For the composer to be able to showcase their “aesthetic sense” and “flair”, there must be a pattern to recognize, so without a reliance on repetition of pitch class or any other parameter than can now continuously be randomized, density and time are of control instead.



Curve functions can be set to any length of time, with adjustable limits and scaling for any parameter of the distribution they are mapped to.

These sort of implementations in Max leave the composer with an infinite number of possibilities with what is possible to randomize and control during a composition. There are tools available that can allow the user to generate notation using these generative systems^[6], with experimentation with graphical scores using Jitter. The inclusion of visuals only further increases the level of abstraction the composer has access to. If desired, the composer can create a graphic score using the same generative techniques, allowing a performer to interpret the music themselves.

With so many possibilities, we no longer need to concern ourselves with the process of creation but the result that comes from such experimentation. Xenakis described processes as these “black box” structures, receiving data and outputting music.^[7] The composition becomes a set of rules, a function, rather than a predetermined piece of music.

Works Cited

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